IPRATROPIUM

Add-on in acute asthma

Protect eyes from nebuliser mist

CHEMICAL NAME (1R,3r,5S,8r)-3-[(2RS)-3-hydroxy-2-phenylpropanoyloxy]-8-methyl-8-(propan-2-yl)-8-azoniabicyclo[3.2.1]octane bromide

CLASS Short-acting muscarinic antagonist (SAMA) bronchodilator

TRADE NAMES Atrovent · Combivent (with salbutamol) · Duovent (with fenoterol) · Rinatec (nasal)

THERAPEUTIC USES / INDICATIONS

- Acute severe asthma (added to salbutamol nebs).
- COPD; symptomatic relief and exacerbations.
- Rhinorrhoea in allergic/non-allergic rhinitis (nasal spray).

ADMINISTRATION

- **Nebulised** with salbutamol (may mix in same chamber).
- **Inhaler (pMDI)** with spacer.
- Use mouthpiece or well-fitting mask — mist in eyes can precipitate angle-closure glaucoma.

DOSAGES

- **Acute asthma (nebulised):** 500 mcg (child 6–12 yr 250 mcg; <6 yr 125–250 mcg) every 20 min for 3 doses, then every 4–6 h.
- **COPD (nebulised):** 250–500 mcg 3–4 times daily.
- **Inhaler:** 20–40 mcg (1–2 puffs) 3–4 times daily.
- **Nasal:** 2 sprays each nostril 2–3 times daily.

PHARMACOKINETICS

- **Onset:** 15 min; peak 1–2 h. **Duration:** 4–6 h.
- Quaternary amine — minimal systemic absorption, does not cross BBB.
- Swallowed fraction excreted in faeces.

PHARMACODYNAMICS

- Blocks **muscarinic (M3) receptors** in airway smooth muscle → ↓ vagal bronchoconstriction and mucus secretion.
- Slower onset than β2 agonists; additive effect.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry mouth.
- Throat irritation, cough, headache.
- Nausea.

ADVERSE EFFECTS (SERIOUS)

- Acute angle-closure glaucoma (mist in eyes).
- Urinary retention (prostatic hypertrophy).
- Paradoxical bronchospasm; tachycardia.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to atropine derivatives.
- **Caution:** angle-closure glaucoma, prostatic hyperplasia, bladder outflow obstruction.

KEY INTERACTIONS

- **Other anticholinergics (tiotropium):** avoid duplicate long-term use.
- **β2 agonists:** additive bronchodilation (beneficial).

NURSING MANAGEMENT

- Assess respiratory status, peak flow, SpO_2.
- Protect eyes; mouthpiece preferred.
- Ask about eye pain, blurred vision, urinary difficulty.
- Oral hygiene for dry mouth.

PATIENT EDUCATION

- Keep the mist away from your eyes.
- Report eye pain or blurred vision, difficulty passing urine.
- Sip water or sugar-free sweets for dry mouth.

KEY POINTS

- SAMA — blocks vagal bronchoconstriction.
- Dry mouth; protect eyes (glaucoma).
- Add-on to salbutamol in acute asthma (500 mcg).
- COPD symptom relief.

SOURCES CHECKED **[453]** Atrovent 250 UDVs, 1 ml (UK SmPC) **[454]** Ipravent CFC-Free Inhaler 20 micrograms per actuation pressurised inhalation, solution (UK SmPC) **[447]** BTS/SIGN asthma guideline (SIGN 158)

Full references p. 432–457